Nucleon circuits

Are the proton and the neutron circuits of light? — register entry H31

kronspice simulation report · 2026-11-07

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Summary

The light-ring picture used in this repository for the electron (H23) and the neutrino (H28) is carried to the nucleons. A light ring is light of energy mc² going once round a ring of radius ħ/(mc); a charge q carried round it at c gives the Dirac moment qħ/2m exactly. Each ring is built as a kronspice inductor with Maxwell mutual coupling to its neighbours. The proton is tried as one ring, the neutron as the rings of its decay products (p + e + $\bar{\nu}$), and the standard accounts — three quark rings, and Skyrme's knot, computed here — are set beside them. **H31 is a hypothesis**; the measured values and the standard models are the yardstick.

- **One light ring is not the proton:** radius 0.2103 fm against the measured 0.84075 fm (4.0×), moment 1.0000 μN against 2.7928 μN (2.79×).
- **The neutron as p + e + $\bar{\nu}$ rings** gets charge 0, spin ½ and instability (+0.7823 MeV) right, but with each ring at its own size its moment is −1835.5 μN, 959× too large; the electron ring that fits −1.9130 μN (0.9897 fm) carries 199.4 MeV against 1.293 MeV available (154×).
- **Three quark rings with one number** (ring energy 335.96 MeV, fixed by μ_p) predict μ_n = −1.8619 μN against the measured −1.9130 (−2.7 %), and electromagnetism makes the neutron 1.053 MeV lighter (lattice QED part −1.00 MeV).
- **Skyrme's knot, computed:** μ_p +1.880, μ_n −1.324, $|\mu_p/\mu_n|$ 1.419 (measured 1.460), radius 0.595 fm (0.72); helium-4 as one B = 4 cube binds 267 MeV (7.7 %) against 28.3 MeV (0.75 %).
- **Lifetime:** the light rings carry no weak coupling, so they give none; the standard model gives 880.5 s against the measured 878.4 s.
- **Status: bounded.** The decay-products picture fails on moment and energy by factors of 150 to 1,000.

What is being tested

Hypothesis H31 (user hypothesis, 2026-11-05): the proton and the neutron are circuits of light, as the electron and the neutrino are pictured in H23 and H28, the neutron being the rings of its decay products. **Standard prediction beside it:** the constituent-quark picture (three spin-weighted quark rings) and Skyrme's knot of the pion field, with the measured nucleon properties any model has to reproduce:

	proton	neutron
charge	+e	0
spin	½	½
mass	938.272 MeV	939.565 MeV (1.293 MeV heavier)
magnetic moment	+2.7928 μN	−1.9130 μN
size	charge radius 0.84075 fm	$\langle r^2 \rangle$ = −0.1155 fm²
lifetime	stable	878.4 s (n → p e $\bar{\nu}$, 0.782 MeV released)

Measured values: CODATA 2022 (masses, moments, r_p); PDG 2024 ($\langle r^2 \rangle_n$, τ_n, V_{ud}, λ). μN is the nuclear magneton.

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The model: a light ring as a circuit

Light of energy mc² going once round a ring of circumference h/(mc) sits at radius R = ħ/(mc). A charge q carried round at c is a current qc/(2πR) on an area πR², so its moment is qcR/2 = qħ/(2m): the Dirac moment, exactly (TestLightRingDirac). In kronspice each ring is an inductor (a thin ring of core radius R/2, L = $\mu_0 R(\ln 16 - 7/4)$) carrying the ring's current from a current source, coupled to its neighbours by Maxwell's coaxial-circle mutual inductance. The ledger books every ½LI² and every $M \cdot I_1 \cdot I_2$, matching the analytic sum to 1e-6 (TestCircuitTwin). The four circuits (one-ring proton, p + e rings, uud and udd) are exported as netlists, KiCad schematics and results sheets in export/.

Results

1. The proton as one light ring

quantity	one light ring	measured
radius ħ/(m_pc)	0.2103 fm	0.84075 fm (4.0× larger)
current	3.635×10^4 A	
moment (kronspice currents)	1.0000 μN	2.7928 μN (2.79× larger)
magnetic self-energy ½LI²	1.1143 MeV (ledger 1.1143 MeV, residual 0.0e+00)	
ring carrying +e at c with the measured moment	radius 0.5874 fm, a light ring of 335.96 MeV	= $m_p c^2/2.7928$

One ring reproduces the moment of a point particle; the measured moment is 2.79 times that. The ring that would carry the measured moment holds about a third of the proton's energy: **one ring cannot be the whole proton.**

2. The neutron as the light rings of its decay products

The neutron decays to a proton, an electron and an antineutrino; in the light model these are three rings — the proton ring, the electron's photon ring (H23) and the inverted neutrino ring (H28).

requirement	result	verdict
charge 0	+e − e + 0 = 0	passes
spin ½	three spin-½ rings can combine to ½ (Rutherford's 1920 bare p + e could not)	passes
unstable	$m_n - (m_p + m_e)$ = +0.7823 MeV: more energy than its parts	passes
moment −1.9130 μN	each ring at its own size (electron ring 386.2 fm): −1835.5 μN, 959× too large	**fails**
moment, refitted	electron ring of radius 0.9897 fm gives −1.9130 μN (kronspice: currents $+1.302\times10^4$ A and −7724 A; magnetic energy 0.1300 MeV, analytic 0.1300, residual 0.0e+00)	fits by construction
energy of that ring	light on a 0.9897 fm ring carries 199.4 MeV against 1.293 MeV available: 154× too much	**fails**
binding	Coulomb attraction of the two rings −1.617 MeV (own Coulomb energies +1.724 MeV)	far too weak

The antineutrino ring is neutral (an E ring: no current, no moment, energy < 0.45 eV), so it changes none of this. The decay-products picture gets the charge, the spin and the instability right, and fails on the moment and the energy by two to three orders of magnitude — the same reason Rutherford's 1920 neutron was abandoned after Chadwick's 1932 discovery.

3. Standard counterpart: three quark light rings

Each quark is a light ring (u: +2/3 e, d: −1/3 e), and the nucleon's spin-½ state is a weighted mix of spin arrangements (for the proton u↑u↑d↓ with weight 2/3, u↑u↓d↑ and u↓u↑d↑ with 1/6 each: the SU(6) wavefunction). Each arrangement is a three-ring kronspice circuit, and its moment is the I·πR² of the solved currents.

	quark rings	measured
quark ring energy (the one free number, from μ_p)	335.96 MeV, radius 0.5874 fm	—
proton moment	+2.7928 μN (fitted)	+2.7928
neutron moment	**−1.8619 μN (predicted)**	−1.9130 (−2.7 %)
μ_n/μ_p	−0.6667 (= −2/3)	−0.6850

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	quark rings	measured
three rings' light energy	1007.9 MeV	938.3 (p) / 939.6 (n): 7.4 % high; binding not in the model
electromagnetic energy (rings 0.587 fm apart)	p 1.7075 MeV, n 0.6547 MeV	
$m_n - m_p$ from electromagnetism	−1.053 MeV; at 0.5R −1.285, 2.0R −0.844, 4.0R −0.688 MeV	lattice QED part −1.00 MeV

The ring that fits the proton's moment in Part 1 is exactly one quark ring; with that one number the neutron's moment follows to 2.7 %. The ring spacing is a declared choice, not fitted. Electromagnetism alone makes the neutron about 1 MeV *lighter*, in line with the lattice's QED part; the measured +1.293 MeV needs the d ring heavier than the u ring (lattice QCD part +2.52 MeV; Borsanyi et al., *Science* 347, 1452, 2015). Ledger residuals 0.0e+00, 0.0e+00.

4. The knot picture: Skyrme's model, computed

In Skyrme's model the proton and the neutron are one knot of the pion field (winding number B = 1) in two states of internal rotation; helium-4 is one knot of winding 4, with cubic symmetry. skyrme/skyrme.py solves the hedgehog profile F(r) (F(0) = π, F(∞) = 0) on a stretched grid r = 1.5·tan(πx/2) of 400 points by L-BFGS-B, in units $1/(eF_\pi)$ for length and F_π/e for energy; the B = 4 cube uses the rational-map ansatz of Houghton, Manton & Sutcliffe (1998), angular integral I = 20.65. The constants e and F_π are fitted to the nucleon and Δ masses, as Adkins, Nappi & Witten (ANW 1983) did.

	this computation	ANW 1983	measured
B = 1 energy (F_π/e)	36.462	36.5	—
e, F_π	5.437, 129.1 MeV	5.45, 129 MeV	F_π = 186 MeV (this convention)
isoscalar radius	0.595 fm	0.59 fm	0.72 fm
μ_p	+1.880 μN	+1.87	+2.793
μ_n	−1.324 μN	−1.31	−1.913
$\lvert\mu_p/\mu_n\rvert$	1.419	1.43	1.460
g_A	not computed	0.61	1.23
B = 4 energy per winding (12π² units)	1.1366	HMS 1.137	—
helium-4 binding	267 MeV (7.7 %)	—	28.3 MeV (0.75 %)

The solver also prints Λ = 50.70 (ANW 50.9), winding B = 1.00005, per-winding energy B = 1 1.2315 (1.232), and classical masses B = 1 866 MeV, B = 4 3196 MeV. The B = 1 knot reproduces ANW to 0.1–1 %. It gives the neutron a negative moment and the moment ratio to 3 % with no charged parts at all, but absolute moments and sizes come out about 30 % low. The cube is lighter per winding than four separate knots, so helium-4 holds together, but about ten times too tightly: the classical knot over-binds, a known shortcoming of the original model.

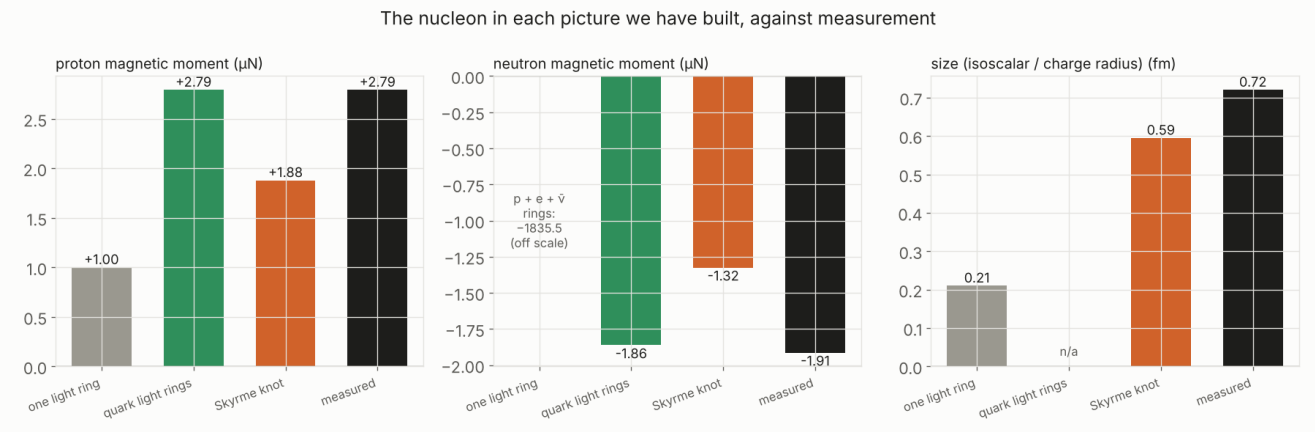


Figure 1. The nucleon in each picture against measurement: proton moment, neutron moment and size for one light ring (grey), quark light rings (green), the Skyrme knot (orange) and measurement (black). The p + e + ν̄ rings' neutron moment, −1835.5 μN, is off the scale. No size is given for the quark rings.

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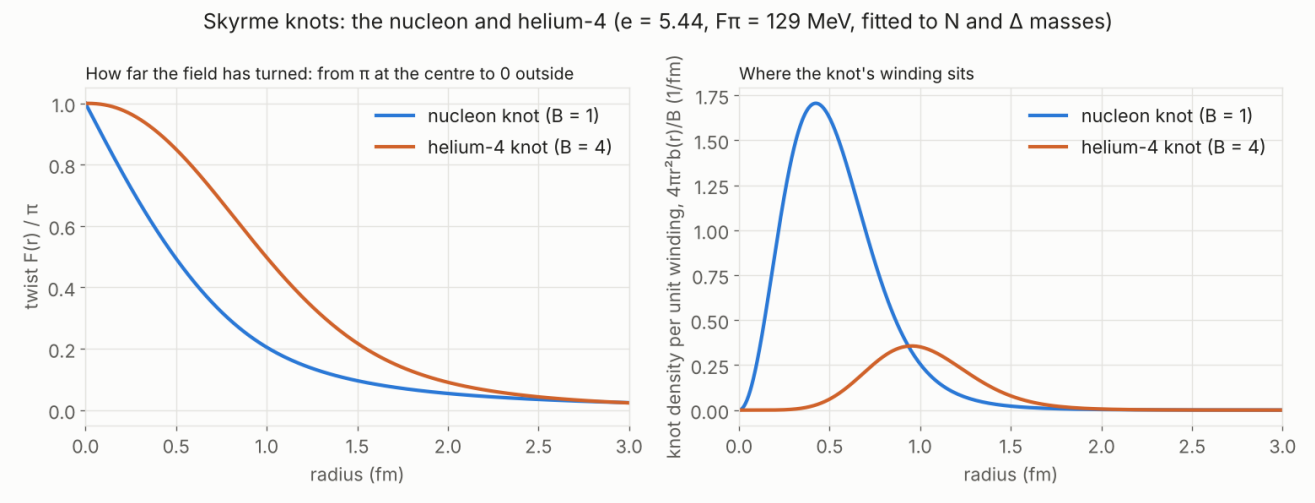


Figure 2. Skyrme knots with e = 5.44, F_π = 129 MeV. Left: the profile F(r)/π, how far the field has turned, from π at the centre to 0 outside, for the nucleon (B = 1) and helium-4 (B = 4). Right: the radial winding density per unit winding, $4\pi r^2 b(r)/B$.

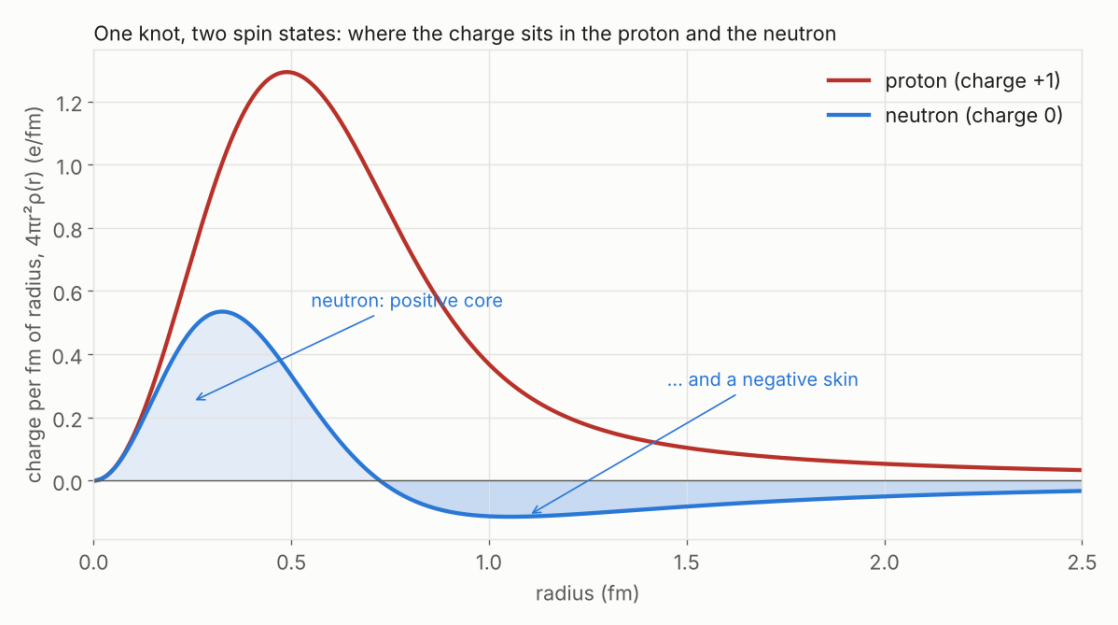


Figure 3. Charge per fm of radius, $4\pi r^2\rho(r)$, in the proton and the neutron, from $\rho = \frac{1}{2}\rho_S \pm \frac{1}{2}\rho_V$ (ρ_S the winding density, ρ_V from the moment of inertia density, each normalized). The neutron has a positive core and a negative skin, the sign pattern electron scattering measures.

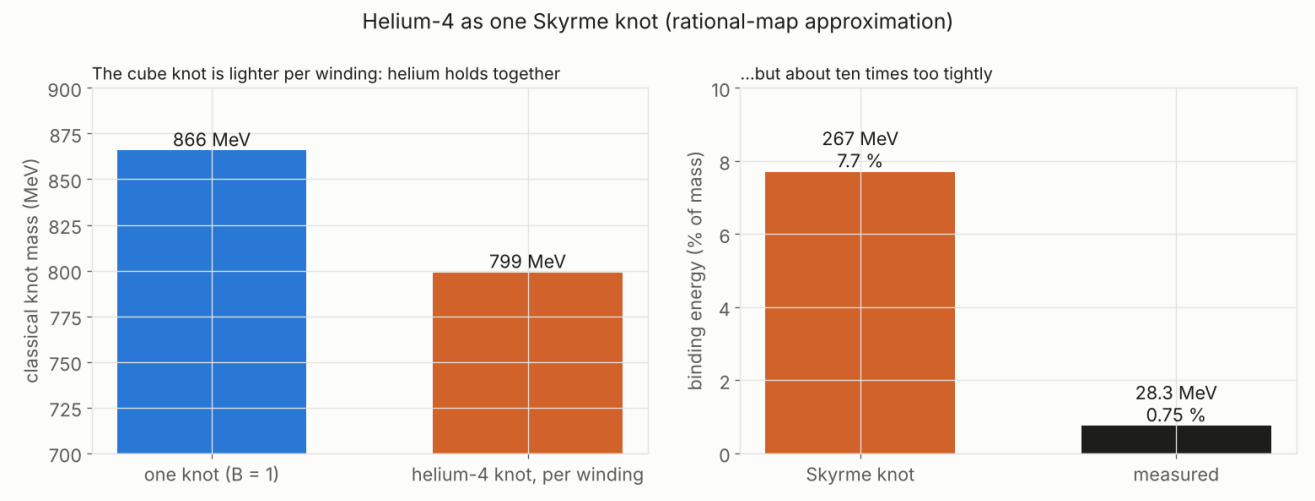


Figure 4. Helium-4 as one Skyrme knot (rational-map approximation). Left: classical knot mass, 866 MeV for one B = 1 knot against 799 MeV per winding in the B = 4 cube, so helium holds together. Right: binding energy as a fraction of mass, 267 MeV (7.7 %) for the knot against the measured 28.3 MeV (0.75 %).

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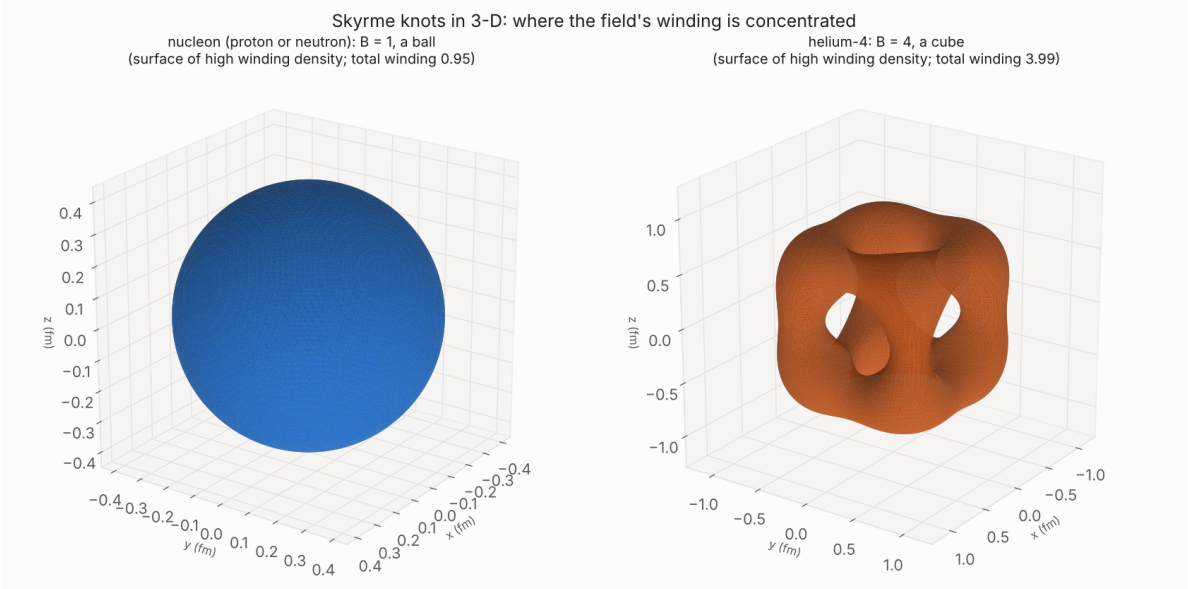


Figure 5. Surfaces of high winding density in 3-D: a ball for the nucleon (B = 1) and a cube for helium-4 (B = 4). The integrated winding inside the plotted box is 0.95 and 3.99.

5. Neutron decay

- **Energy budget:** 939.5654 − 938.2721 − 0.5110 = 0.7823 MeV, shared by the electron and the antineutrino.
- **Lifetime:** the light rings carry no weak coupling, so the light model gives no lifetime. The standard model gives $\tau = 4908.6\ \text{s}/(V_{ud}^2(1 + 3\lambda^2)) = 880.5$ s with $V_{ud} = 0.97373$ and $\lambda = 1.2754$ (Czarnecki, Marciano & Sirlin, *PRL* 120, 202002, 2018), against the measured 878.4 s.

What it means

- The light ring captures one thing exactly: a ring of light carrying a charge has the Dirac moment $q\hbar/2m$.
- The proton is not one ring, and the neutron is not its decay products held together: that picture fails on moment and energy by factors of 150 to 1,000.
- Three rings work: the quark-ring circuit predicts the neutron's moment to 2.7 % from one number, and gets the electromagnetic part of the mass difference in sign and size.
- The knot (Skyrme) gets the same pattern with no charged parts.
- Neither the light model nor the circuits supply the binding that holds three rings together (in the standard account, gluons) or the weak interaction that turns d into u in the decay.

Validation

test	checks	held to
TestLightRingDirac	ring of radius $\hbar/(m_p c)$ with +e at c: moment 1 μN from the circuit's current; radius 0.2103 fm	1e-9; 1e-6
TestCircuitTwin	three coupled rings, both current signs: ledger stored energy = ½ΣLI² + $\Sigma M I_1 I_2$; currents as set; ledger	1e-6; 1e-9
TestQuarkRings	μ_p fitted, $\mu_n/\mu_p = -2/3$; electromagnetic $m_n - m_p$ near −1 MeV	1e-9; ±0.5 MeV
TestNeutronFromDecayRings	electron ring for the measured moment 0.990 fm; its energy > 100× the 1.293 MeV available	1e-3
TestLifetime	standard-model τ within 0.5 % of 878.4 s	0.5 %
TestExport	the four ring circuits as netlists, KiCad schematics and results sheets	—

Limits of the model

- Rings are thin coaxial rings with a declared core radius R/2 and declared spacings; the spacing of the quark rings is chosen, not fitted.

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- No binding force and no weak interaction in the ring circuits.
- The Skyrme knot is classical (no quantized vibrations), and the B = 4 cube is the rational-map approximation.

Status

H31: bounded. One ring gives the proton 1 μN (measured 2.793) at 0.210 fm (0.841); the neutron as p + e + $\bar{\nu}$ rings gets charge 0, spin ½ and instability right but its moment 959× too large, and the electron ring that fits the moment carries 154× the available energy. The standard counterparts — quark rings (μ_n −1.862 from one number) and Skyrme's knot — are computed beside it, and the lifetime comes only from the standard weak interaction.

Reproducing the results

```
go run ./examples/nucleon_circuits      # under a second
go test ./examples/nucleon_circuits
KRONSPICE_EXPORT=1 go test ./examples/nucleon_circuits   # export/: netlists, KiCad, results sheets
cd examples/nucleon_circuits/skyrme
python3 skyrme.py      # solves both knots, writes skyrme.json (a few seconds)
python3 render.py      # fig1-fig5 from skyrme.json
```


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